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Real-Time Data Ingestion Using Spark Structured Streaming & PostgreSQL

Project Summary

In this project, you'll build a real-time data pipeline that simulates an e-commerce platform tracking user activity.

You'll generate fake user events (like product views and purchases), stream them using Apache Spark Structured Streaming, and store the processed data in a PostgreSQL database.

You'll gain practical experience with:

- Real-time data streaming
- Data transformation using Spark
- Writing to a relational database
- Handling continuously arriving data

Learning Objectives

By the end of this project, you will be able to:

- Simulate and ingest streaming data
- Use Spark Structured Streaming to process data in real time
- Store and verify processed data in a PostgreSQL database
- Understand the architecture of a real-time data pipeline
- Measure and evaluate system performance

Tools & Technologies ?

- **Apache Spark Structured Streaming**
- **PostgreSQL**
- **Python** (for data generation)
- **SQL** (for database setup)

Project Structure

Part 1: Simulate Data

- Write a script that creates fake e-commerce events (CSV files).
- These events include user actions like "view" or "purchase", along with product info and timestamps.

Part 2: Stream with Spark

- Use Spark Structured Streaming to monitor the folder where CSVs are saved.
- Read and process new CSV files as they appear.
- Apply any needed transformations (e.g., data cleaning, type conversion).

Part 3: Store in PostgreSQL

- Set up a PostgreSQL database and create a table to store incoming events.
- Connect Spark to the database.
- Insert processed data into the table in real time.

Deliverables

You will submit the following:

Deliverable	Description
`data_generator.py`	Python script to generate CSV event data
`spark_streaming_to_postgres.py`	Spark Structured Streaming job to process and write data
`postgres_setup.sql`	SQL script to create database and table
`postgres_connection_details.txt`	Text file with host, port, user, and password
`project_overview.md`	Short write-up explaining your system's components and flow
`user_guide.md`	Step-by-step instructions on how to run your project
`test_cases.md`	Manual test plan with expected vs actual outcomes
`performance_metrics.md`	Report with system performance data (latency, throughput, etc.)
`system_architecture.png`	Diagram showing data flow and components

What to Test

- Are the CSV files being generated correctly?
- Is Spark detecting and processing new files?
- Are the data transformations correct?
- Is data being written into PostgreSQL without errors?
- Are performance metrics (like processing speed) within expected limits?

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